

DATA STRUCTURES AND ALGORITHMS – 3 (25CS2103E)

PLACEMENT HUB

2520030128 – R C Sai Sharan

2520030083 – K.Saanvi

2520030487 – S. Ketana Sindhu

Under the Esteemed Guidance of

Dr. V.Sireesha

Sec 6

CSE

CONTENTS

• Abstract	03
• Introduction	04
• Literature survey	05
• Problem Statement and Objectives	06
• Hardware and Software requirements	07 08
• Results	09
• Conclusion	10
• Future Scope	11

ABSTRACT

Placement Hub is a placement management system that brings student, company, job, eligibility, and application information into one platform. Unlike basic placement systems, it uses DSA as a core part of the system for searching, sorting, HashMap-based retrieval, and matching. It provides matching scores and job recommendations based on skills, academics, and company criteria, helping students find suitable opportunities more efficiently.

INTRODUCTION

- Managing campus placements manually becomes difficult when student and company data increases. Students may also find it difficult to identify jobs that match their skills and eligibility.
- Placement Hub provides a centralized solution where students can manage their profiles, search jobs, check eligibility, and track applications. Its key feature is DSA-based matching and ranking, which goes beyond basic eligibility checking to identify suitable student-company combinations.

LITERATURE SURVEY

- Traditional placement systems mainly provide basic features such as student registration, company management, job posting, and application tracking.
- Searching and sorting may be available, but they are not always the core part of the system.
- Many systems focus mainly on basic eligibility rather than detailed matching.
- HashMap-based retrieval and DSA-based processing are rarely highlighted as major features.
- Placement Hub focuses on matching scores, recommendations, and ranking logic using DSA techniques.

PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

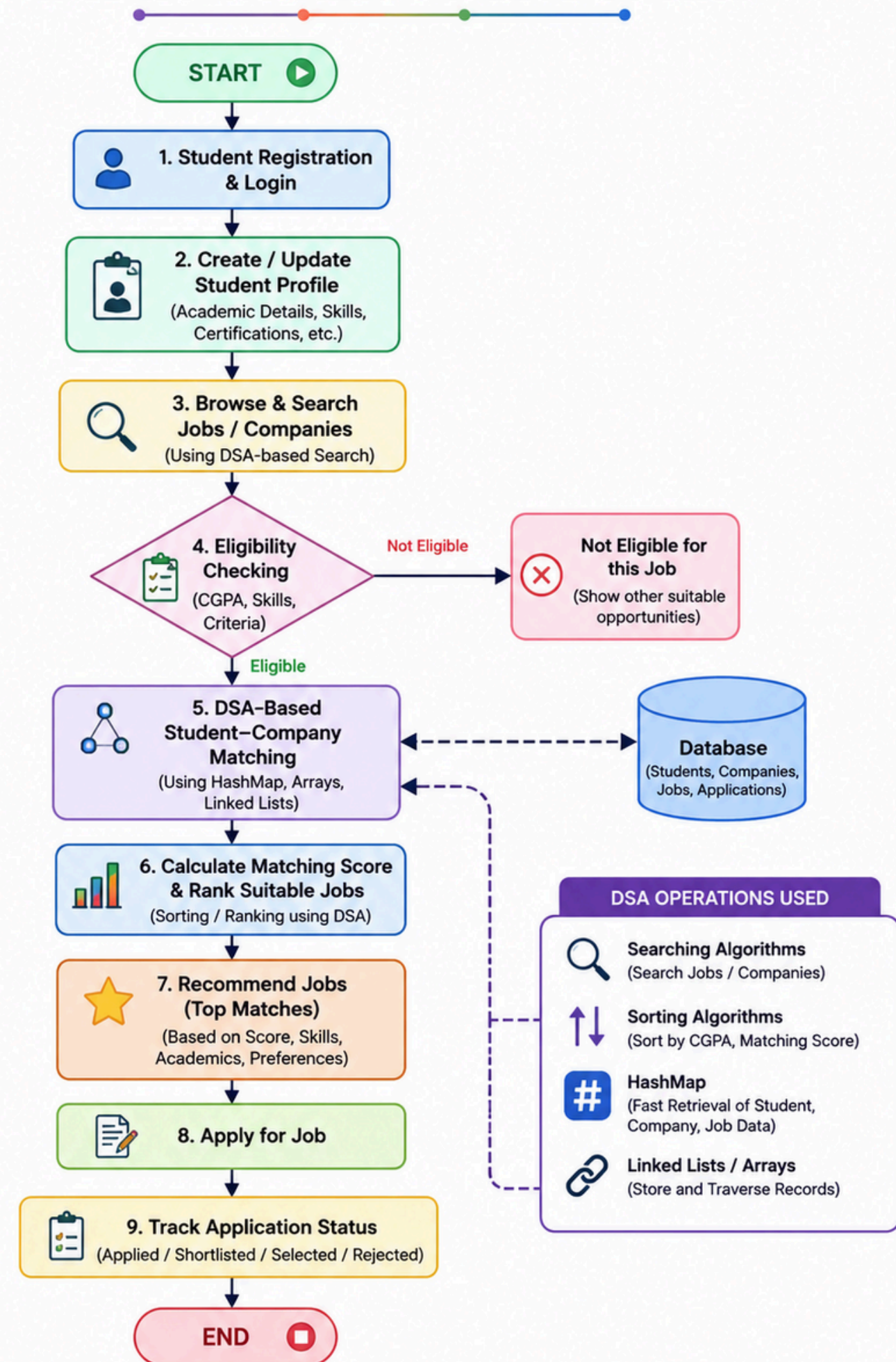
Traditional placement management can make it difficult to handle large amounts of student and company data. Basic systems may only check eligibility and do not effectively rank or recommend the most suitable opportunities.

Objectives

- Centralize student, company, and job information.
- Provide quick and efficient data retrieval.
- Check student eligibility for different jobs.
- Calculate student-company matching scores.
- Recommend jobs based on skills, academics, and criteria.
- Rank suitable opportunities using DSA-based logic.
- Reduce manual work in placement management.

FLOW CHART

PLACEMENT HUB – FLOW CHART



RESULTS

The proposed system provides:

- Faster data retrieval using HashMap.
- Efficient searching and sorting of placement records.
- Quick eligibility checking.
- Matching scores for student-company compatibility.
- Job recommendations based on skills, academics, and criteria.
- Ranking of suitable opportunities.
- Reduced manual effort for the placement process.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements

- Computer / Laptop
- Minimum 4 GB RAM
- Minimum 20 GB free storage
- Internet connection

Software Requirements

- Programming Language: Java
- Database: MySQL
- Frontend: HTML, CSS, JavaScript
- Backend: Java with JDBC
- DSA: Arrays, Linked Lists, Searching, Sorting, HashMap

CONCLUSION

Placement Hub provides an organized and efficient solution for campus placement management. Its major advantage is that DSA is used as a core part of the system, rather than only for demonstration. Searching, sorting, HashMap retrieval, matching scores, and ranking logic help the system provide better student-company matching and job recommendations while reducing manual effort.

FUTURE SCOPE

- **AI-Based Recommendations:** Improve job recommendations by analyzing student skills, academics, and previous applications.
- **Mobile Application:** Provide students and placement cells with easy access through smartphones.
- **Real-Time Notifications:** Notify students about new job postings, application deadlines, interviews, and selection updates.
- **Advanced Matching:** Add more parameters to improve matching scores and ranking accuracy.



THANK YOU